

NOVAC Internship – Task 10 Daily Documentation

Task: Document Naming System Integration, AI-Generated Chunk Heading Improvements, Multi-Chunk Contextual Retrieval Enhancement, Conversational Retrieval Optimization, and Chatbot Interaction Workflow Refinement

Overview

Today's work focused on improving the overall usability, organization, and conversational intelligence of the Retrieval-Augmented Generation (RAG) chatbot platform.

The primary objective was to redesign the document handling and contextual retrieval workflows to make the platform behave more like a scalable enterprise knowledge-base assistant rather than a basic semantic search chatbot.

Major improvements were implemented in three core areas:

- document organization and storage workflows
- chunk readability and retrieval navigation
- conversational multi-context retrieval handling

The system was upgraded to support user-defined document names instead of relying on raw uploaded filenames, enabling cleaner knowledge-base management and preventing filename-based retrieval conflicts.

Additionally, semantic chunk visualization was improved by integrating AI-generated chunk titles and contextual subheadings for every stored chunk, significantly improving readability and chunk accessibility.

The semantic retrieval architecture was also upgraded from single-chunk retrieval to multi-chunk contextual retrieval, allowing the chatbot to combine multiple relevant semantic chunks before generating conversational responses through Mistral AI.

By the end of today's implementation, the platform successfully supported:

- custom document naming workflows
- AI-generated chunk headings
- cleaner chunk visualization
- improved chunk navigation
- multi-chunk contextual retrieval
- enhanced conversational grounding
- improved response depth
- context-aware semantic reasoning
- retrieval aggregation workflows
- conversational UX improvements
- enter-to-send chatbot interaction
- production-style retrieval orchestration

Resources and References Used

- FastAPI Documentation
- MongoDB Documentation
- Groq API Documentation
- Mistral AI Documentation
- Sentence Transformers Documentation
- Semantic Search Concepts
- Retrieval-Augmented Generation (RAG) Concepts
- Vector Similarity Retrieval Concepts
- Conversational AI Architecture Concepts
- Prompt Engineering Concepts

Concepts Learned

Custom Document Naming Systems

Learned how enterprise-grade knowledge-base systems avoid relying directly on uploaded filenames for document management.

Implemented:

- custom user-defined document naming
- overwrite-safe document workflows
- MongoDB document identity refinement
- original filename preservation

Understood:

- scalable document organization
- retrieval-safe naming systems
- enterprise knowledge-base structuring
- document lifecycle management

AI-Generated Chunk Headings

Improved chunk readability by integrating semantic subheadings generated through Groq AI.

Implemented:

- AI-generated chunk titles
- semantic chunk labeling
- contextual subheading generation
- retrieval visualization improvements

Understood:

- semantic navigation systems
- retrieval traceability
- knowledge-base readability optimization
- contextual chunk indexing

Multi-Chunk Contextual Retrieval

Upgraded the chatbot from single-chunk retrieval to multi-context retrieval orchestration.

Previous retrieval workflow:

single chunk retrieval
isolated semantic context
limited response depth

New retrieval workflow:

top-k semantic retrieval
multi-chunk context aggregation
combined contextual grounding
improved conversational reasoning

Implemented:

- top semantic chunk selection
- contextual aggregation pipelines
- multi-context prompt injection

- conversational grounding workflows

Understood:

- retrieval aggregation systems
- context orchestration pipelines
- semantic context fusion
- conversational retrieval reasoning

Conversational UX Improvements

Improved the frontend chatbot interaction experience.

Implemented:

- enter-to-send functionality
- cleaner upload workflows
- document naming input system
- retrieval visualization improvements

Understood:

- conversational UX optimization
- messaging-style interaction systems
- chatbot usability refinement
- frontend interaction workflows

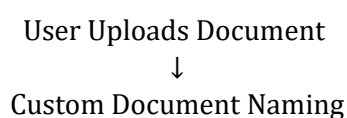
Libraries, Modules, and Components Learned

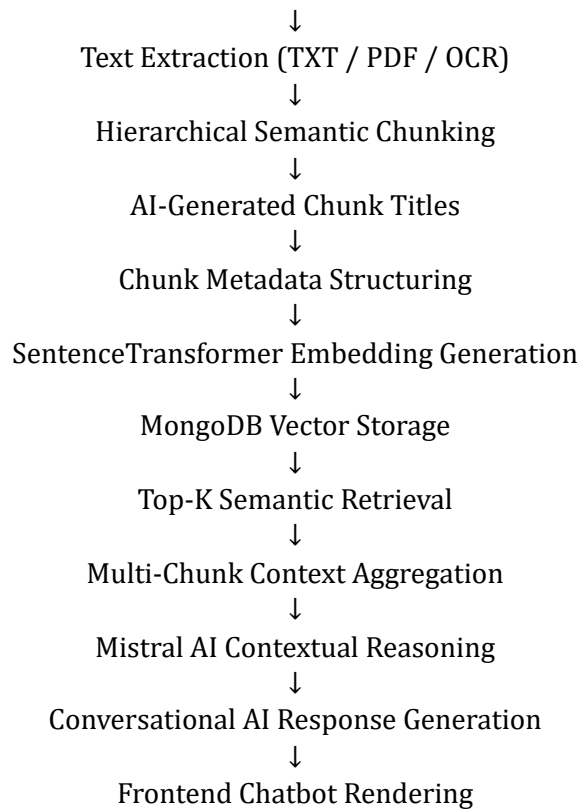
- **FastAPI Form Handling:** Used for custom document name handling during uploads.
- **MongoDB:** Used for document persistence and chunk management.
- **Sentence Transformers:** Used for semantic embedding generation.
- **Groq SDK:** Used for semantic chunk title generation.
- **Mistral AI SDK:** Used for conversational contextual response generation.
- **cosine_similarity:** Used for semantic similarity ranking.
- **React State Management:** Used for frontend document naming workflows.
- **FormData API:** Used for multipart upload handling.
- **JSON Workflows:** Used for semantic chunk metadata handling.

Practical Implementation

- custom document naming architecture
- AI-generated chunk title workflows
- semantic chunk metadata storage
- MongoDB document schema refinement
- multi-chunk retrieval orchestration
- contextual prompt aggregation
- conversational retrieval enhancement
- improved chunk card rendering
- retrieval visualization improvements
- frontend upload workflow refinement
- enter-to-send chatbot interaction
- semantic context fusion workflows
- retrieval-oriented conversational grounding

Architecture Understood





Challenges Faced

Filename Collision Problems

Encountered:

- duplicate filename conflicts
- poor knowledge-base organization
- inconsistent document tracking

Resolved by:

- custom document naming workflows
- MongoDB schema refinement
- overwrite-safe storage handling

Weak Contextual Responses

Previous chatbot behavior produced:

- isolated contextual answers
- shallow semantic grounding
- limited retrieval coverage

Resolved through:

- multi-chunk contextual retrieval
- top-k semantic aggregation
- contextual prompt fusion

Chunk Readability Problems

Initial chunk display behavior:

- difficult chunk navigation

- poor readability
- weak contextual identification

Resolved by:

- AI-generated chunk titles
- semantic subheadings
- improved chunk card organization

Chatbot Interaction Limitations

Encountered:

- manual send button dependency
- inconsistent interaction workflow

Resolved by:

- enter-to-send functionality
- messaging-style chatbot interaction handling

Conclusion

Today's work significantly improved the platform's usability, conversational intelligence, and retrieval orchestration architecture.

The chatbot now behaves much closer to a production-style enterprise RAG assistant with:

- structured document organization
- AI-generated semantic chunk labeling
- contextual retrieval aggregation
- multi-context conversational grounding
- improved retrieval depth
- cleaner semantic navigation
- scalable knowledge-base workflows
- enhanced conversational UX

The concepts implemented today form an important foundation for:

- enterprise AI assistants
- conversational knowledge-base systems
- production-grade RAG architectures
- contextual retrieval systems
- semantic document orchestration
- intelligent retrieval workflows
- scalable conversational AI platforms

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